

Embodied Co-Reflection: Enhancing Fitness Data Reflection through the Integration of Embodied Interaction and Narrative Feedback

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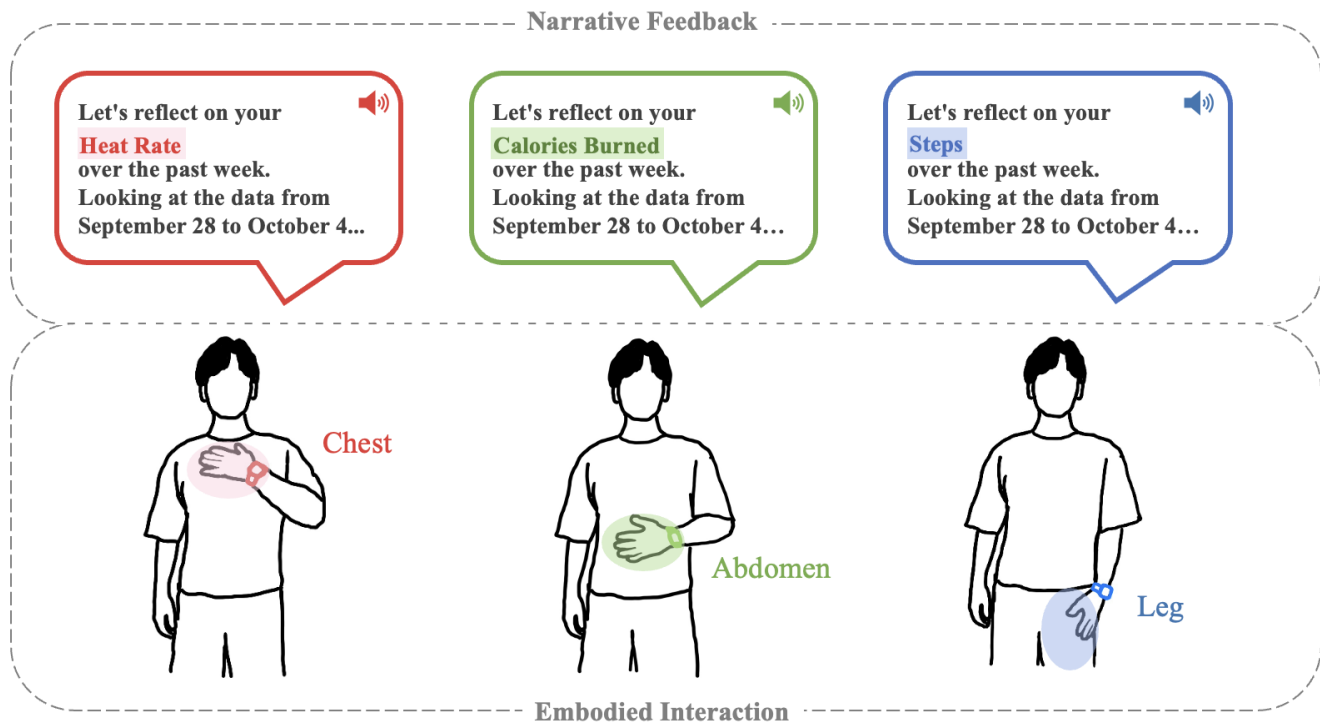


Figure 1: System Interaction Diagram: This diagram illustrates the interaction between users and the system to receive feedback on fitness data. Gestures directed at specific body parts (the chest for heart rate, the abdomen for calories burned, and the leg for steps) trigger voice narrative corresponding to each data type.

Abstract

Many fitness trackers monitor physical activity by presenting data such as heart rate and step count in numerical or graphical formats. However, these methods often fail to connect the data with users' subjective bodily sensations, making it challenging for users to perceive the information as personally relevant. To address this issue, this study proposes a reflective system that integrates embodied

interaction with voice narrative feedback. The system utilizes the body as a reference frame for fitness data and presents health information in a story-like format. Results from user studies indicate that the proposed system significantly improves engagement (aesthetic appeal, reward factors) and reduces cognitive load (satisfaction, frustration), showing moderate to large effect sizes. It also strengthens the connection between users and their data. This approach demonstrates the potential to overcome the limitations of traditional fitness trackers, offering a novel and immersive reflective experience in the healthcare and fitness domains.

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CCS Concepts

• **Human-centered computing** → **Interaction paradigms**; *User interface design*; *Gestural input*; *Interaction devices*.

Keywords

Embodied Interaction, Fitness Data Reflection, Large Language Models, Voice Feedback, User Engagement, Human-Computer Interaction

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1 INTRODUCTION

Physical activity is vital for extending healthy life expectancy and reducing the risk of lifestyle-related diseases. The WHO recommends 150–300 minutes of moderate-to-vigorous exercise per week [8]. However, a 2022 report indicates that approximately one-third of adults worldwide do not meet these recommended levels, making physical inactivity a critical issue [26]. Recently, wearable devices—such as fitness trackers have rapidly gained popularity as a means to address this shortfall. These devices measure and visualize quantitative data such as heart rate, step count, and calories burned, supporting goal setting, self-monitoring, and encouraging behavioral change [6, 10, 18, 21]. However, existing tools primarily rely on visual presentation of numerical data and graphs, which have been criticized for their limitations in allowing users to connect the data with their subjective bodily sensations [14, 22]. Users are required to consciously interpret the presented information and mentally associate it with themselves, which increases cognitive load and makes it more challenging to intuitively perceive the information as personally relevant [2, 29]. From the perspective of Embodied Interaction, the body is not merely a means of input but rather the foundation for data interpretation and self-awareness, underscoring the importance of seamlessly integrating data with the user's bodily experience [11]. In this research, we propose a method that leverages this Embodied Interaction approach while incorporating a narrative component into the feedback. Narrative can enrich quantitative data by grounding it in users' life context and emotions, thereby enhancing their intrinsic motivation to change behavior [25]. To evaluate the effectiveness of our proposed method, we conducted two small-scale preliminary studies. The results suggest that employing narrative in the voice modality as feedback, within an Embodied Interaction context, can significantly increase user engagement and foster a sense of connection between the user's body and the data. Furthermore, combining Embodied Interaction with voice-based narrative appears to reduce cognitive load during system operation compared to simple display-based interactions, potentially enhancing data comprehension. This study highlights the potential of this approach to improve the user experience in health management and fitness contexts.

2 RELATED WORK

2.1 Personal Informatics

Personal Informatics has been widely studied as a framework for collecting users' behavioral and physiological data to facilitate

reflection and behavioral change [16, 24]. Li et al. proposed a five-stage model—data collection, integration, reflection, and behavior change—which serves as the basis for self-monitoring [17]. Specifically, recording daily activities fosters a sense of “awareness,” leading to a stage where users can understand the causes and context of their behaviors through introspection on the data. Subsequently, users create an action plan based on the insights gained and implement it, thereby achieving behavioral change. However, this model does not assume that all users aim for behavioral change; some focus primarily on deepening their understanding of data, omitting the behavioral stage [30]. To address this diverse needs, supporting the reflection process becomes crucial [4]. Particularly, designs that encourage active reflection—such as interactive systems [15] are suggested to have the potential to enhance user engagement.

2.2 Embodied Interaction

Traditional fitness trackers have predominantly employed a visual-centric approach, displaying numeric data and graphs on screens. However, this method has been criticized for failing to effectively link the data to users' bodily sensations, making intuitive understanding difficult [9]. This issue stems from the well-known limitation of the cognitive interaction model in the Human-Computer Interaction field [29], which often separates mind and body, and treating users as information processors and thereby neglecting bodily experiences and interactions with the environment [1]. As an alternative, Embodied Interaction, based on embodiment theory, has gained attention. This theory posits that cognition emerges through the interaction of the body, environment, and brain, emphasizing the importance of positioning the user's body as the basis for data interpretation [3]. For example, “FEEL” [5] provides sensory feedback to the soles of the feet to encourage unconscious behavior change in reducing shower time, illustrating how leveraging bodily sensations can naturally guide user actions. In the fitness domain, “In Your Face!” [23] employs AR technology to project fitness tracker data directly onto the user's face, aiming to foster emotional and intuitive connections with the data. Additionally, “Magic Mirror” [28] proposes a method that uses the body as the reference frame for data, associating metrics such as heart rate and step count with specific body parts through gestures. However, these approaches confine the role of embodied interaction to merely triggering for data referencing, creating a disconnect between data selection and its interpretation. Specifically, Magic Mirror's reliance on displaying information through screens may hinder the integration of data with the self, potentially forcing users into “deliberate cognition.” From the perspective of the D4EB [29] design principles, this indicates that the system does not fully utilize sensorimotor coupling, falling short of achieving a truly embodied interaction.

2.3 Narrative Feedback

As a potential solution to the shortcomings of existing visual-centric fitness trackers, this study focuses on narrative feedback through voice. Narrative feedback presents data not as mere numbers, but as contextualized, story-like information connected to users' daily lives and emotions. This approach deepens data understanding and promotes behavioral change [25]. In Rooksby et al.'s research shows

that linking health data to real-life contexts enhance users' motivation for behavior change [24]. More recently, advancements in large language models (LLMs) have opened up new possibilities for more flexible narrative generation that adapts to user contexts. Strömél et al. demonstrated that LLM-based narrative feedback can improve user concentration and satisfaction compared to purely visual numeric presentations [27].

This study proposes a method to address the limitations of existing systems, particularly the issue of "disconnection between data and the self caused by information presentation through displays." The approach involves presenting data triggered by embodied interaction as voice-based narrative feedback. According to Mayer's "modality effect," presenting information in audio format can distribute cognitive load across both visual and auditory channels, thereby reducing overall cognitive burden [19]. By integrating embodied interaction with voice-based narrative feedback, this study aims to develop a novel reflection system that enhances the user's sense of connection with their data.

3 EMBODIED CO-REFLECTION

3.1 System Design

Our system is designed based on body schema theory [7], enabling users to perform gesture operations referencing specific body parts and intuitively acquire fitness data. In contrast, the conventional cognitive interaction model primarily relies on visual information presentation through displays, often resulting in a disconnect between the user and the data. By leveraging embodied interaction and providing voice-based narrative feedback, this system aims to address this issue, making the information more relatable and personally relevant to the user (Figure 2). According to Hummels et al. [13], physical movements complement cognitive understanding and enable users to engage more deeply with their environment and data. In this system, gestures corresponding to data such as heart rate, calories burned, and steps are selected to promote a natural connection between the user's bodily sensations and the data. Based on the study by Subramonyam [28], intuitive correspondences are used, such as associating the chest with heart rate, the abdomen with calories burned, and the leg with steps. Additionally, narrative feedback is generated according to the study by Strömél et al. [27], providing concise and neutral summaries of fitness data trends without relying on graphs or numerical displays. Furthermore, according to the modality effect described by Mayer et al. [19], presenting information through both auditory and visual channels distributes information processing across different sensory modalities, reducing cognitive load. In this system, voice feedback is utilized to allow users to focus more on their bodily sensations without relying on visual information, thereby enhancing their immersion in the fitness data.

3.2 System Implementation

The system developed in this study leverages the Apple Watch to enable users to intuitively retrieve fitness data through gestures and receive feedback based on that data. The system's main functions are categorized into three components: data collection, gesture recognition, and feedback generation. First, the system collects fitness data, such as heart rate, calories burned, and step count

for the past week, using the HealthKit API. Next, it utilizes an accelerometer and gyroscope to recognize users' gestures in real time. Gesture detection involves comparing the arm angles calibrated in the recording mode with the current angles obtained in the detection mode to trigger data corresponding to the body part being touched. Once a gesture is recognized, the system generates narrative feedback based on the retrieved data using the LLM (GPT-4). The text generated by the LLM is then converted into voice using a Text-to-Speech API. This sequential process allows users to conveniently and intuitively retrieve fitness data through gestures, linking the data to their bodily sensations for effective reflection. Below is an example of feedback generated by the LLM:

"Let's reflect on your heart rate over the past week. Looking at the data from January 1, fluctuations in heart rate were observed throughout the week. On the first day, the values were relatively stable, but the following days saw increased variability, with a temporary peak in the higher range on January 5. Midweek, further changes were observed, with the upper limit of heart rate reaching higher values. This data suggests that your heart rate patterns are diverse and influenced by the intensity and circumstances of your activities. Overall, your heart rate has shown active fluctuations throughout the week, reflecting responses to specific training or activities."

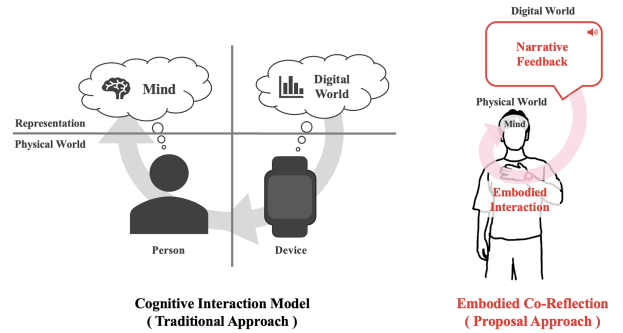


Figure 2: Comparison between the Cognitive Interaction Model and the Embodied Co-Reflection System. The traditional approach relies on information presentation through displays, often creating a disconnect between the self and the data. In contrast, the proposed method leverages embodied interaction and provides narrative feedback through voice, making it easier for users to perceive the presented information as personally relevant.

4 EVALUATION

4.1 Study1: Comparison of Feedback Modalities

4.1.1 Overview. The aim of Study 1 was to investigate how different feedback modalities affect user engagement and bodily perception in an Embodied Interaction setting designed for reflecting on fitness data. Using data collected from a fitness tracker, two feedback modalities were compared:

- (1) Text Narrative: Visual information presentation via a display
- (2) Voice Narrative: Non-visual information presentation without a display

The influence of these modalities on the user experience was examined through both quantitative and qualitative evaluations. The experiment involved 10 engineering students (aged 21–25 years, 5 with prior reflection experience), and a crossover design was adopted to eliminate order effects. Participants selected fitness data using gestures and received feedback based on the selected data. After each session, engagement was evaluated using the UES-SF [20], which measures dimensions such as focused attention (FA), perceived usability (PU), aesthetic appeal (AE), and reward factor (RW), on a 7-point Likert scale. Additionally, participants were interviewed to gather insights about system usability, preferences for feedback modalities, and their perceived connection to their bodies.

4.1.2 Results. Quantitative evaluation was conducted using the Wilcoxon signed-rank test. The results showed that aesthetic appeal (AE) scores were significantly higher for voice narrative ($p = 0.007$), with a large effect size ($r = 0.85$). For the reward factor (RW), while no significant difference was observed ($p = 0.07$), a moderate effect size ($r = 0.57$) was suggested. The quantitative results are illustrated in the box plot shown in (Figure 3). From the interviews, 8 out of 10 participants reported feeling a stronger bodily connection with voice narrative, and all participants found voice narrative to be more approachable. Ultimately, 7 participants expressed a preference for voice narrative. Qualitative feedback indicated that voice feedback facilitated intuitive and quick understanding. However, participants also highlighted challenges, such as difficulty hearing in outdoor settings and inconvenience when comparing long-term data (P3, P5, P6). Many participants noted that voice narrative naturally aligned with gestures without relying on visual information, enhancing their perceived connection to the data (P1, P3, P9). In contrast, text narrative was considered suitable for detailed data verification and numerical comparisons but was described as mechanical and impersonal by some participants (P6, P7, P10).

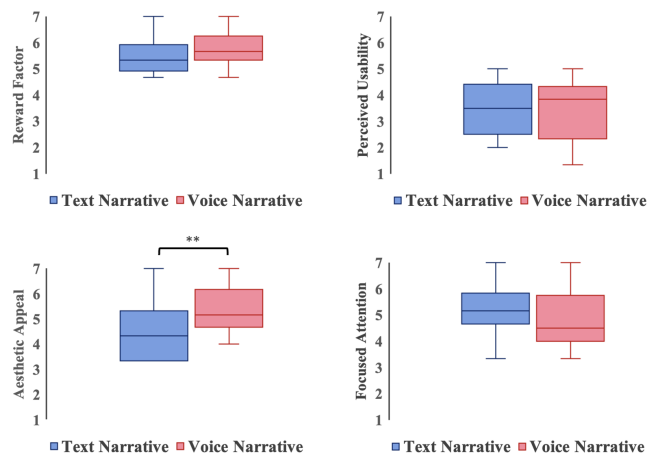


Figure 3: Subscale scores of UES-SF (indicates a significant difference between text and voice at $p < 0.01$)**

4.2 Study2: Effects of Embodied Interaction

4.2.1 Overview. The purpose of Study 2 was to investigate how a reflection support system that integrates Embodied Interaction and voice narrative influences cognitive load and bodily perception during reflection. Using data collected from a fitness tracker, we compared the following two interaction modalities:

- (1) Tap Operation: Voice narrative presented via a display
- (2) Gesture Operation: Voice narrative presented without a display

The impact of these modalities on user experience was examined through both quantitative and qualitative evaluations. The experiment involved 10 engineering students (aged 21–24 years, 6 with prior reflection experience). Each participant selected fitness data using one of the interaction methods and received voice feedback based on the selected data. A crossover design was adopted, and after each session, participants evaluated cognitive load using the NASA-TLX [12] on a 7-point Likert scale, measuring dimensions such as physical demand, mental demand, temporal demand, performance, effort, and frustration. Additionally, oral interviews were conducted to gather insights into bodily sensations during feedback.

4.2.2 Results. Quantitative evaluation was conducted using the Wilcoxon signed-rank test. The results showed that Gesture Interaction resulted in significantly lower frustration scores compared to Tap Interaction ($p = 0.04$), with a large effect size ($r = 0.65$). For Performance, while no significant difference was observed ($p = 0.20$), a moderate effect size ($r = 0.41$) was indicated. The quantitative results are illustrated in the box plot in (Figure 4). In the interviews, 9 out of 10 participants reported feeling a stronger connection between the data and their body with gesture interaction compared to tap interaction. The interviews revealed that most participants strongly experienced a sense of bodily connection to the target data and found gesture interaction to be more intuitive. They also evaluated it as enabling them to better understand the data as personally relevant compared to tap interaction. For instance, one participant noted that placing their hand on their chest strengthened the connection between their body and heart rate data (P2), while another reported feeling that the voice feedback was more personal as it responded to their physical actions (P3). In contrast, participants noted that tap interaction, while convenient, often led to reduced engagement, as it involved simply listening passively after the interaction (P4, P9).

5 DISCUSSION

This study examined the impact of a feedback system that integrates voice narrative and embodied interaction on user experience through two experiments. In Experiment 1, a comparison of text narrative and voice narrative as feedback modalities revealed that voice narrative significantly enhanced aesthetic appeal and had a moderate effect on the reward factor. These results suggest that voice feedback may transcend simple information delivery and be perceived as a valuable process that is naturally and intuitively accepted by users. Furthermore, interviews indicated that voice narrative fosters a sense of familiarity and intuitively connect with bodily sensations, thereby enhancing user immersion.

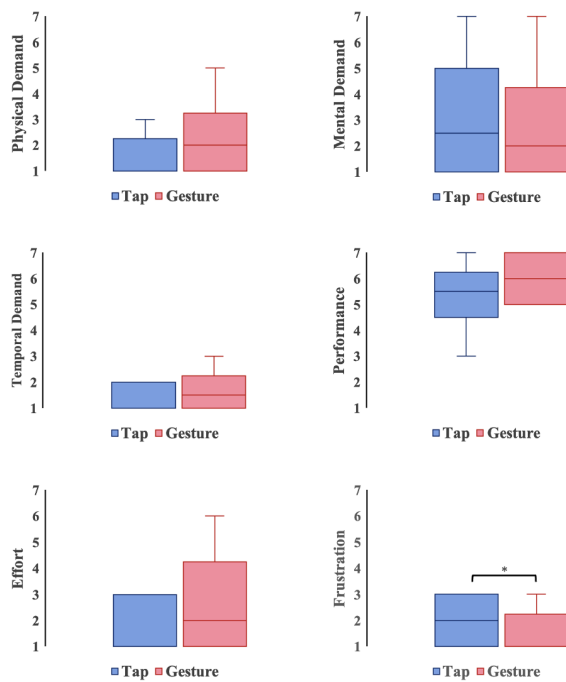


Figure 4: Subscale scores of NASA-TLX (* indicates a significant difference between Tap and Gesture at $p < 0.05$)

In Experiment 2, two interaction methods—gesture-based interaction and tap-based interaction—were compared to evaluate their effects on cognitive load and bodily awareness. Gesture-based interaction demonstrated significantly lower cognitive load compared to tap-based interaction in the frustration category and showed a moderate effect size on satisfaction. Furthermore, participants reported feeling a stronger connection between the data and their bodies through gesture-based interaction, suggesting that it not only improves usability but also provides a novel experience where users perceive the data as personally relevant. While tap-based interaction was appreciated for its simplicity, it was observed to lead to more passive engagement after the initial interaction. In summary, the combination of voice narrative and gesture-based interaction is considered an effective approach to enhancing intuitive data comprehension and self-relevance by leveraging users' bodily awareness. However, to effectively utilize these characteristics, it is necessary to address practical challenges, such as noisy environments and the comparison of multiple datasets, through careful system design. Additionally, this study was conducted with a small sample of university students, highlighting the need for further long-term evaluations across diverse user groups.

6 CONCLUSION

This study demonstrates the potential to address the challenges of traditional vision-dependent fitness trackers by combining embodied interaction with voice-based narrative feedback. The experimental results revealed that the combination of voice-based narrative feedback and embodied interaction produced medium to

large effect sizes in enhancing engagement (aesthetic appeal, reward factors) and reducing cognitive load (satisfaction, frustration). Additionally, participants' comments indicated that the proposed system strengthens the connection between the data and their bodily sensations, helping them perceive the presented information as personally relevant. These findings suggest that the system bridges the gap between abstract fitness data and users' bodily experiences, promoting a more intuitive and immersive reflective experience. Future work should include long-term evaluations involving a more diverse range of user groups and scenarios to verify the system's practical applicability. Moreover, integrating other modalities, such as visual and tactile feedback through hybrid designs, holds promise for sustaining behavior change and intrinsic motivation.

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